HR MANAGEMENT SYSTEM

Project Report

Technology: Spring Boot • Java • MySQL

Prepared for project submission

1. Project Overview

The HR Management System is a REST API application developed with Spring Boot and MySQL. It provides a structured backend for managing employee-related information and leave requests. The application exposes REST endpoints that can be tested through Swagger UI.

2. Objectives

- Provide REST endpoints for employee and related HR operations.
- Support create, read, update, and delete (CRUD) operations.
- Validate incoming request data and return useful error responses.
- Use a relational database to persist application data.
- Apply pagination, sorting, search queries, and transaction management where implemented.
- Demonstrate security, auditing, logging, caching, monitoring, and database connection pooling.
- Provide automated test coverage for selected application components.

3. Technology Stack

Technology	Purpose
Java	Application programming language (runtime used for final JAR: Java 26 in the verified run).
Spring Boot 4.1.1	Application setup and embedded server integration.
Spring Web / REST	HTTP endpoints and REST responses.
Spring Data JPA / Hibernate	Database persistence and ORM.
MySQL 8.0.46	Relational database.
HikariCP	Database connection pooling.
Spring Security	Authentication and role-based authorization.
Springdoc OpenAPI / Swagger UI	Interactive API documentation and manual endpoint testing.
Spring Boot Actuator	Application health endpoint.
Maven	Dependency management, testing, and packaging.

4. Main Features

- Employee CRUD operations through REST endpoints.
- Request validation and centralized exception handling.
- Pagination and sorting for employee data.
- Employee search using JPQL and a native SQL query.
- Leave-request operations with transaction management.
- Audit fields for creation and modification metadata.
- Application logging and scheduled task support.
- Caching configuration.
- HTTP Basic security with demonstration roles for administration, HR, and employees.
- Swagger UI and Actuator health monitoring.

- HikariCP database connection pool.

5. High-Level Architecture

Client / Swagger UI → REST Controller → Service Layer → Repository Layer → MySQL. Cross-cutting components include validation, exception handling, security, auditing, logging, caching, transactions, scheduling, Actuator, and HikariCP.

6. Database Setup

The database used by the application is named **hr_management**. The project includes a SQL export at **database/hr_management.sql**. Import this file into MySQL Workbench or another MySQL client, and ensure the MySQL server is running before starting the application.

Database connection settings are supplied through environment variables rather than a hard-coded password. Configure **DB_USERNAME** and **DB_PASSWORD** in the environment used to launch the application. Do not put real database credentials into GitHub, the report, or the submission ZIP.

7. Running the Application

- Install a compatible Java runtime and ensure MySQL is running.
- Create or import the hr_management database using database/hr_management.sql.
- Set DB_USERNAME and DB_PASSWORD in the environment.
- From the project root, run: `mvn clean package`.
- Run the generated JAR: `java -jar target/hrmanagement-0.0.1-SNAPSHOT.jar`.
- Open Swagger UI at http://localhost:8080/swagger-ui/index.html.
- Check application health at http://localhost:8080/actuator/health.

8. Security Notes

The project uses HTTP Basic authentication and in-memory demonstration accounts for ADMIN, HR, and EMPLOYEE roles. These credentials are for demonstration only and should be changed or replaced with an appropriate user-management approach before any real deployment. Do not publish production credentials.

9. Testing and Verification

The recorded Maven test run completed successfully with 17 tests, 0 failures, 0 errors, and 0 skipped. Manual verification included CRUD operations through Swagger UI, a successful employee update response (HTTP 200), a successful employee deletion response (HTTP 204), Actuator health reporting UP, and HikariCP startup logging showing HRManagementHikariPool - Start completed.

Screenshots are stored in the screenshots directory. The HikariCP startup log demonstrates successful pool initialization; it is not a load or performance stress test.

10. Project Deliverables

- Java source code and automated tests.
- MySQL database export: database/hr_management.sql.
- README.md project setup and usage instructions.
- Screenshots of API operations, health, security, startup, and test results.
- Executable JAR: target/hrmanagement-0.0.1-SNAPSHOT.jar (submit separately if required).
- This project report.

11. Repository

GitHub: https://github.com/Hrithikmgowda/HRManagementSystem

12. Conclusion

The HR Management System demonstrates a layered Spring Boot REST application backed by MySQL, with employee and leave-related functionality and supporting capabilities such as validation, security, auditing, logging, caching, monitoring, transactions, and connection pooling. The project was packaged as an executable JAR and its recorded test suite passed.